

Question	Marking Guidance	Mark	Comments
02.1	Accept any two of the following for one mark: Concentration/mass of yeast Concentration/mass of glucose/sugar (solution/s) Concentration of oxygen (in solution) Volume of yeast Volume of glucose/sugar (solution/s) pH (of glucose/sugar solution/s) Species/type of yeast Size/diameter/volume of test tubes	1 (AO3)	Ignore number of yeast cells Ignore amount for volume or concentration

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02.2	Correct answer of $2.5 \times 10^{-2} = 2 \text{ marks};;$ 4.2×10^{-2} (scale read from the wrong direction) = 1 mark OR 2.5×10^{-3} (scale read as 3.7 mm not 3.7 cm) = 1 mark OR 2.4×10^{-2} (scale read as 36.5) = 1 mark OR Correct answer not in standard form and/or not to 2 significant figures, eg 0.0247/0.025/ 2.47×10^{-2} = 1 mark OR Answer in incorrect standard form eg 25×10^{-3} or $0.25 \times 10^{-1} = 1 \text{ mark}$ OR Evidence of $37 \div 1500$ or $37 \div (25 \times 60) = 1 \text{ mark}$	2 (2 x AO2)	

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02.3	1. Maltose is a disaccharide; 2. (Time to be) hydrolysed into glucose OR (Time to) break glycosidic bond(s);	2 (2 x AO2)	1. Accept maltose is made of two glucose/ monosaccharides 2. Accept monosaccharides for glucose

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02.4	This method would be less accurate because 1. Colour/endpoint is subjective; The accuracy of this method could be improved by 2. (Have a) colour standard (of the yellow) as a reference for the end point; OR (Have a) colorimeter/absorbance/transmission reading (of the yellow) as a reference for the end point;	2 (2 x AO3)	1. Accept descriptions of the word 'subjective' 2. Accept use a colorimeter to time how long it takes to reach a certain value/absorbance/transmission 2. Ignore any references to coloured filters used in a colorimeter

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02.5	All 4 correct = 2 marks;; Any 2 or 3 correct = 1 mark <pre> graph TD Pyruvate[Pyruvate] --> Box1[] Pyruvate --> ReducedNAD[Reduced NAD] Box1 --> AcetylCoA[Acetylcoenzyme A] CoenzymeA[Coenzyme A] --> Box1 </pre>	2 (2 x AO1)	Accept pyruvic acid for pyruvate For reduced NAD accept NADH, NADH + H⁺ or NADH₂ For reduced NAD reject reduced NADP / NADPH / NADPH₂ / NADPH + H⁺ For acetylcoenzyme A accept acetyl co A
